Two, Not One, Electrophysiological Correlates of Consciousness: Evidence From a No-report Inattentional Blindness Paradigm

[1]Nicolacoudis, Angelica Maria; [2]Bruzadin Nunes, Ugo; [3]Sarig, Adi; [2]Fish, Nicholas; [3]Mudrik, Liad; [1]Pitts, Michael; [2]Schurger, Aaron

[1]Department of Psychology, Reed College; [2]Department of Psychology, Brain Institute, Chapman University; [3]Sagol School of Neuroscience, Tel Aviv University

SCALPLAB
Sensation Cognition Attention Language Perception

Introduction

- We use Inattentional Blindness (IB) [1] in the context of a "no-report" paradigm [2] to investigate neural correlates of consciousness (NCCs).
- 3-phase IB paradigms allow for isolation NCCs from task-specific activity [3].

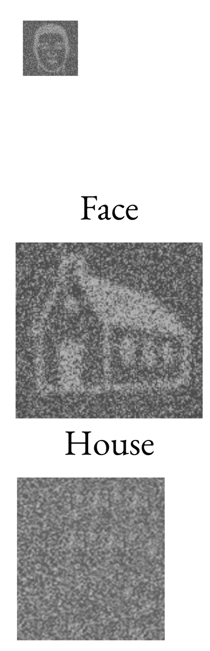
- We used four different analyses of scalp EEG data to help inform this debate:

1) Event-related potentials (ERPs) 2) Event-related spectral perturbations (ERSPs)
3) Signal stability analyses 4) Pattern classification with temporal generalization

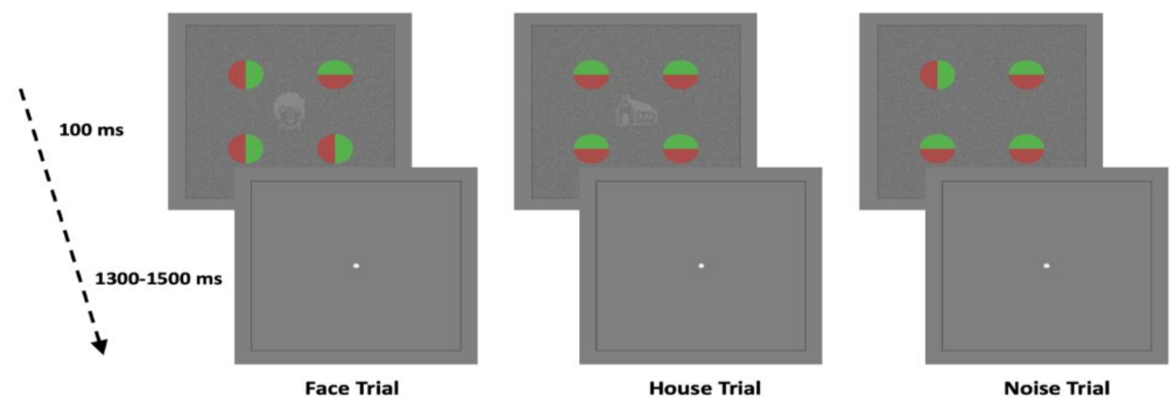
Methods

Design

Stimuli:



Stimulus Sequence:



Experiment Sequence:

Phase I	Phase II	Phase III
Task: Attend to colored circles, compare orientation (same/diff)	Task: Attend to colored circles, compare orientation (same/diff)	Task: Attend to critical stimuli, discriminate faces, houses, noise
Inattentional blindness to stimuli	Aware of task-irrelevant stimuli	Aware of task-relevant stimuli

Awareness Assessment (between Phase I and Phase II); Awareness Assessment (between Phase II and Phase III)

Data from two labs:
Reed (N=42)
Tel Aviv (N=30)

~45% of participants inattentionally blind

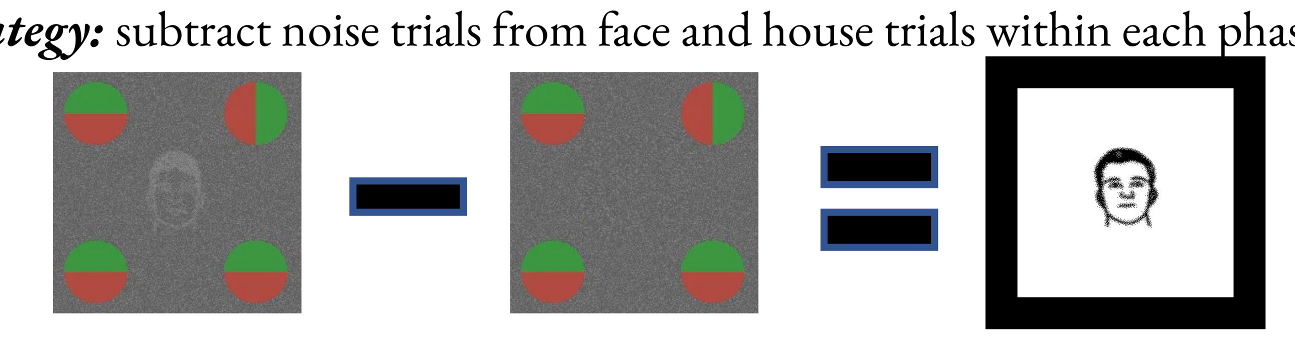
# of trials per phase:

- 66.6% stims, 33.3% blanks
- 288 trials of each type (faces, houses, blanks)
- Stims physically identical across phases

Analysis

Goal: identify differences in brain activity between noticing (aware) and not noticing (blind) face and house stimuli

Strategy: subtract noise trials from face and house trials within each phase:

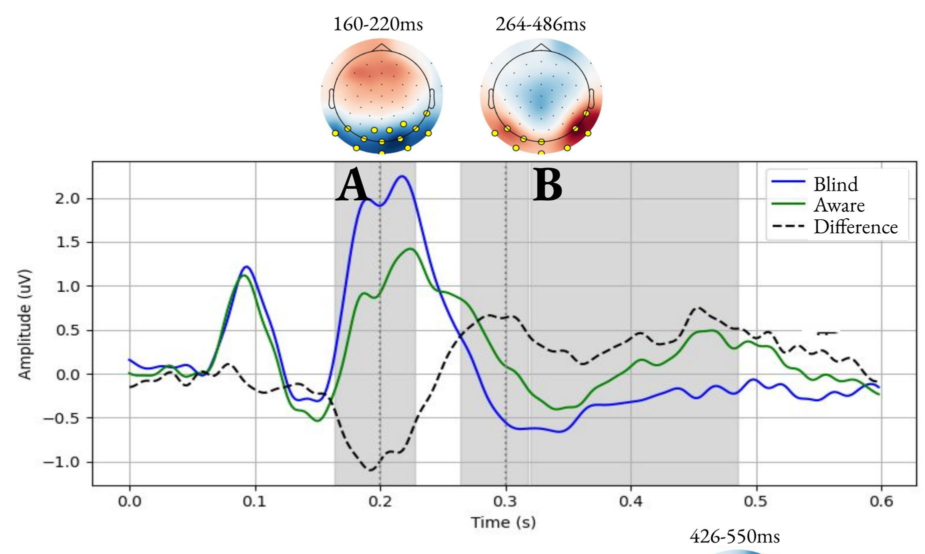


Main Contrast: Phase 2 (aware, task irrelevant) vs. Phase 1 (blind, task irrelevant)

ERPs

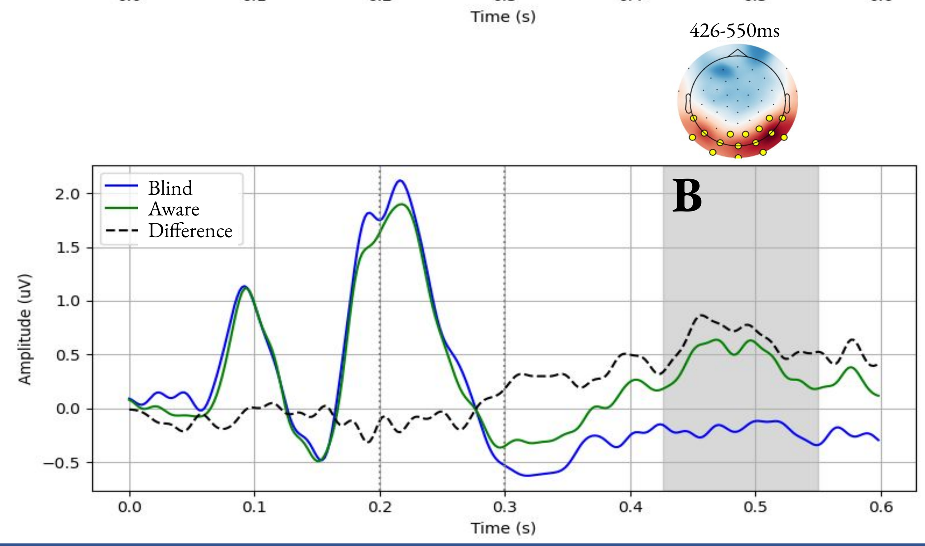
Faces (Aware – Blind)

A: N170/VAN
B: Late visual awareness positivity (LVAP)



Houses (Aware – Blind)

A: early signal not significant
B: Late visual awareness positivity (LVAP)



ERSPs

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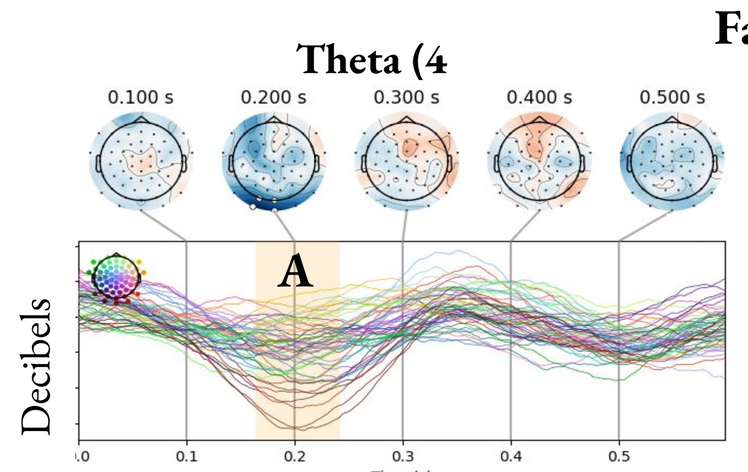
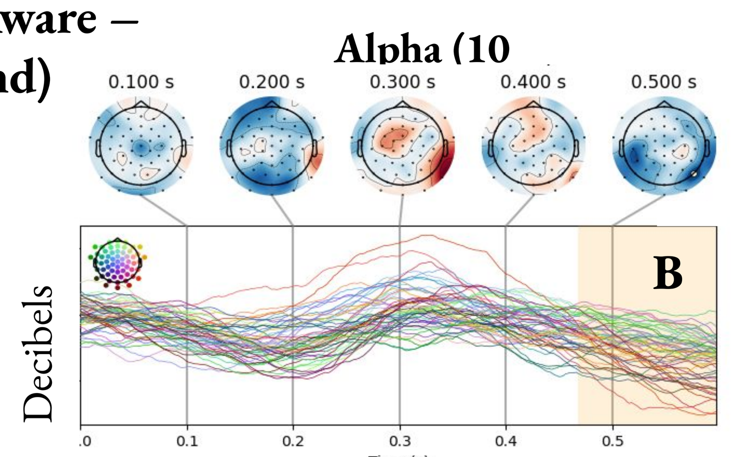
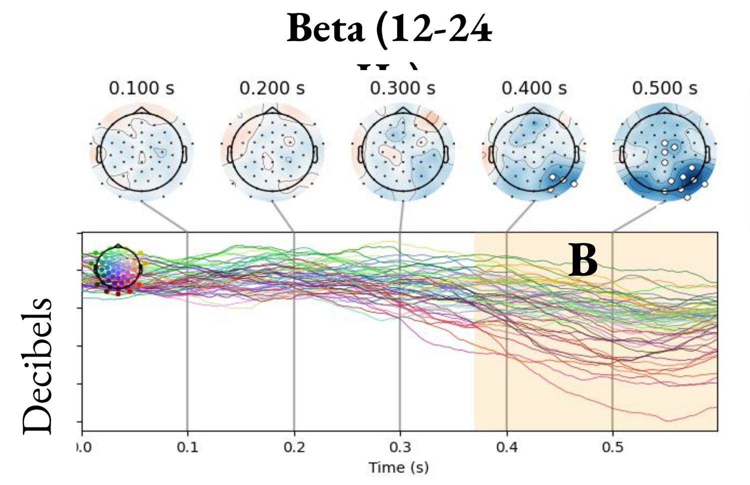
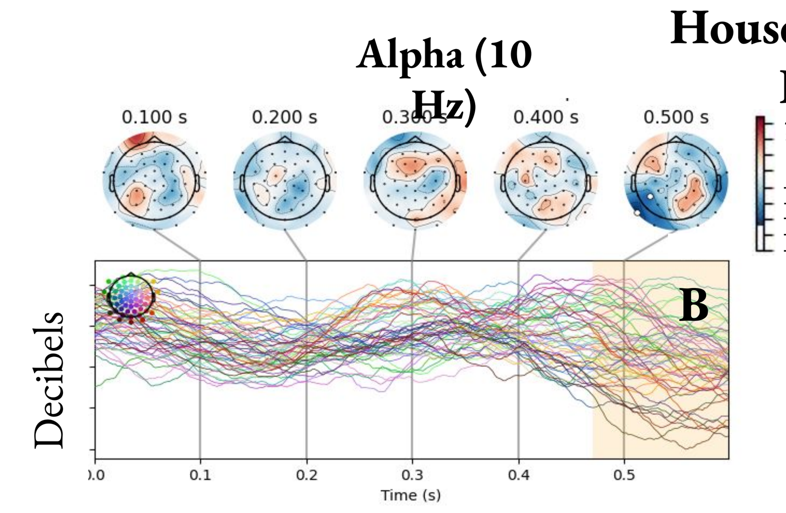


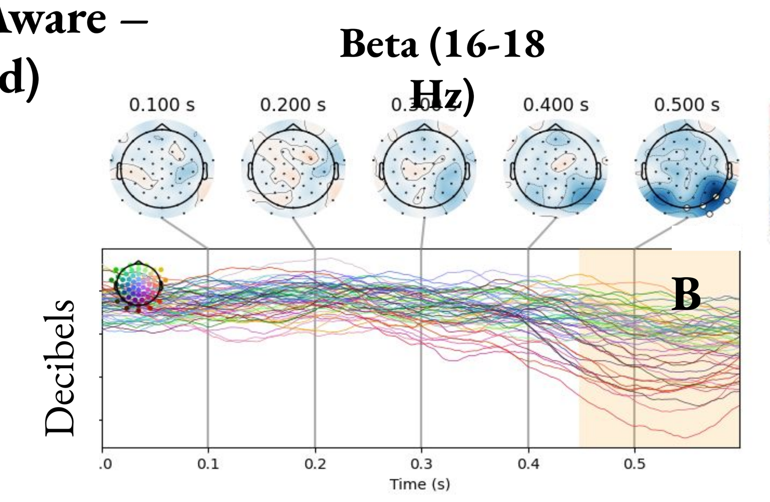




A: Early theta activity
B: Late alpha and beta activity

Houses (Aware – Blind)

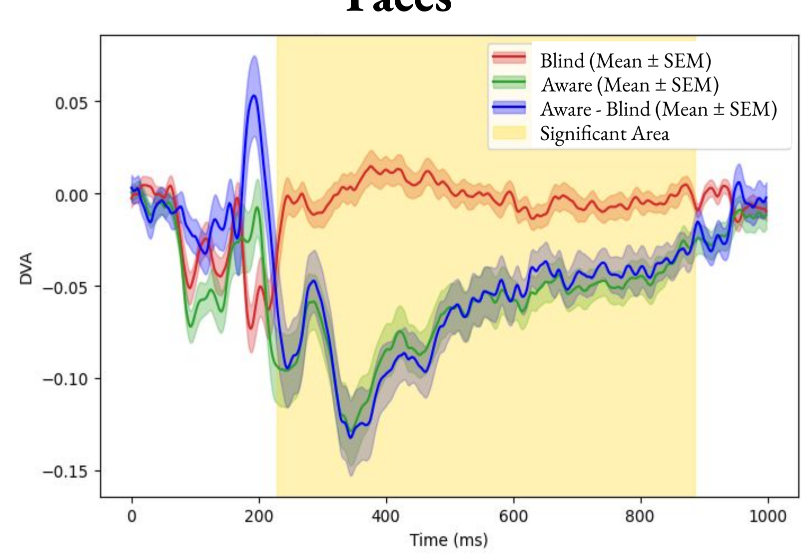




A: No significant early theta activity
B: Late alpha and beta activity

Pattern Stability (Directional Variance)

Faces



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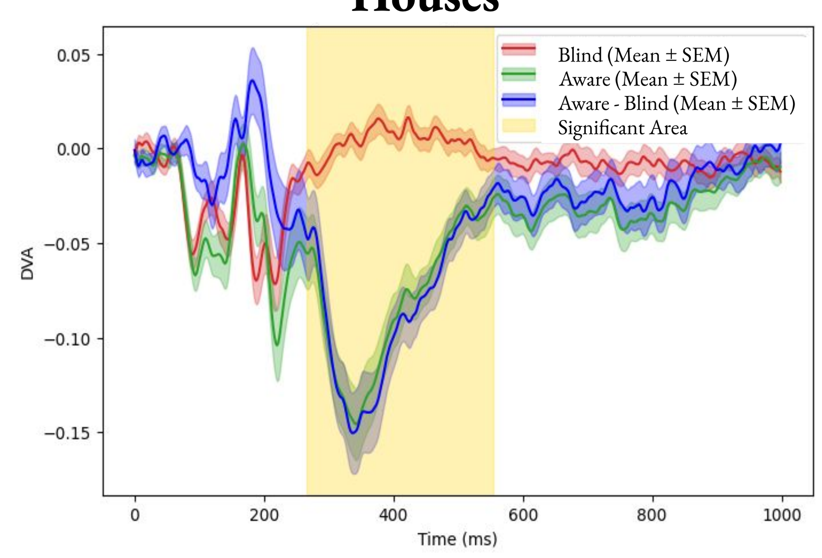
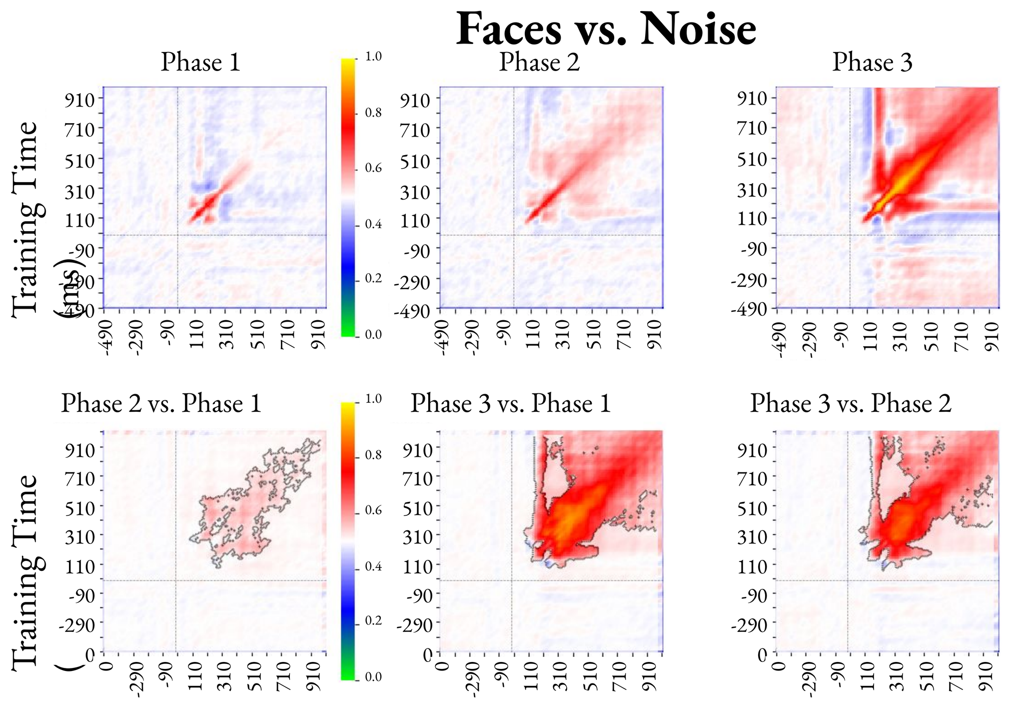


Temporal Generalization of Decoding

Faces vs. Noise



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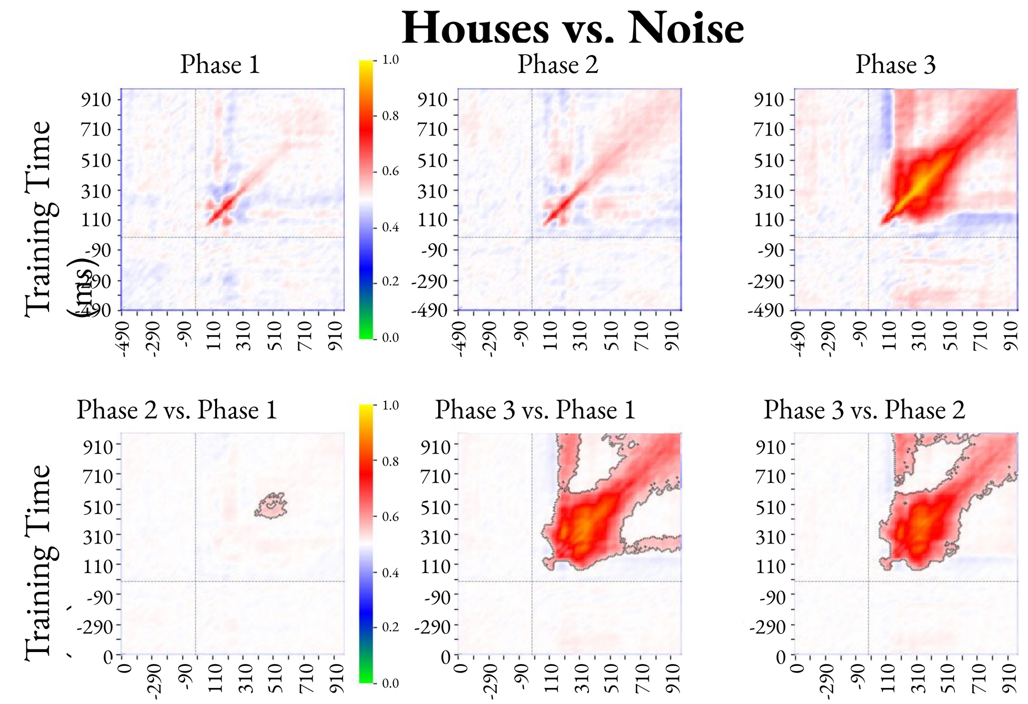
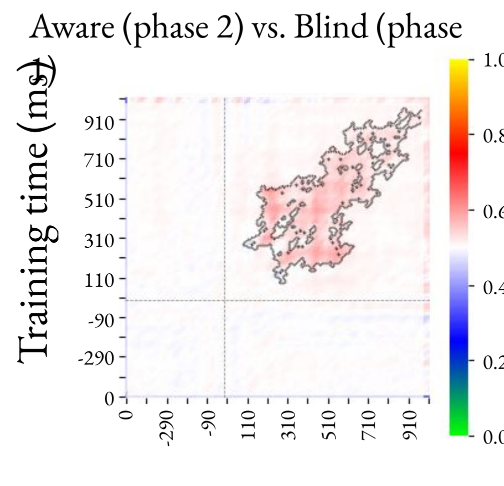


Faces

Aware (phase 2) vs. Blind (phase 1)



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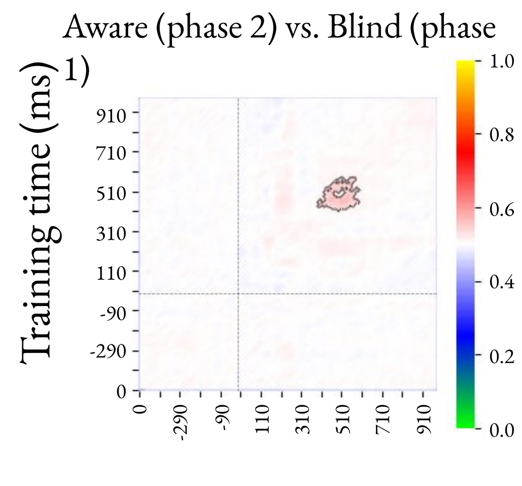

Discussion

- Possible early signal (A) of awareness, which may be stimulus specific (N170) or just stronger for faces (VAN)
- Late signal (B) was present for both stimuli, and may be more generally related to awareness:
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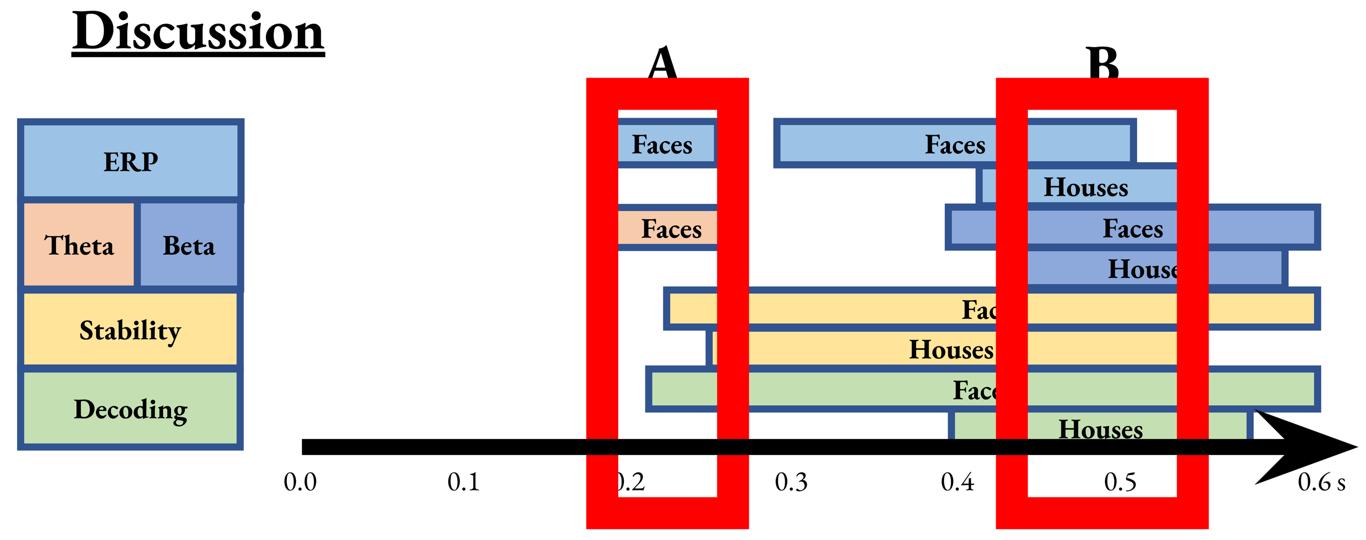
  - Results replicated in 30 participants in an independent lab (Tel Aviv University)



References

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Acknowledgements

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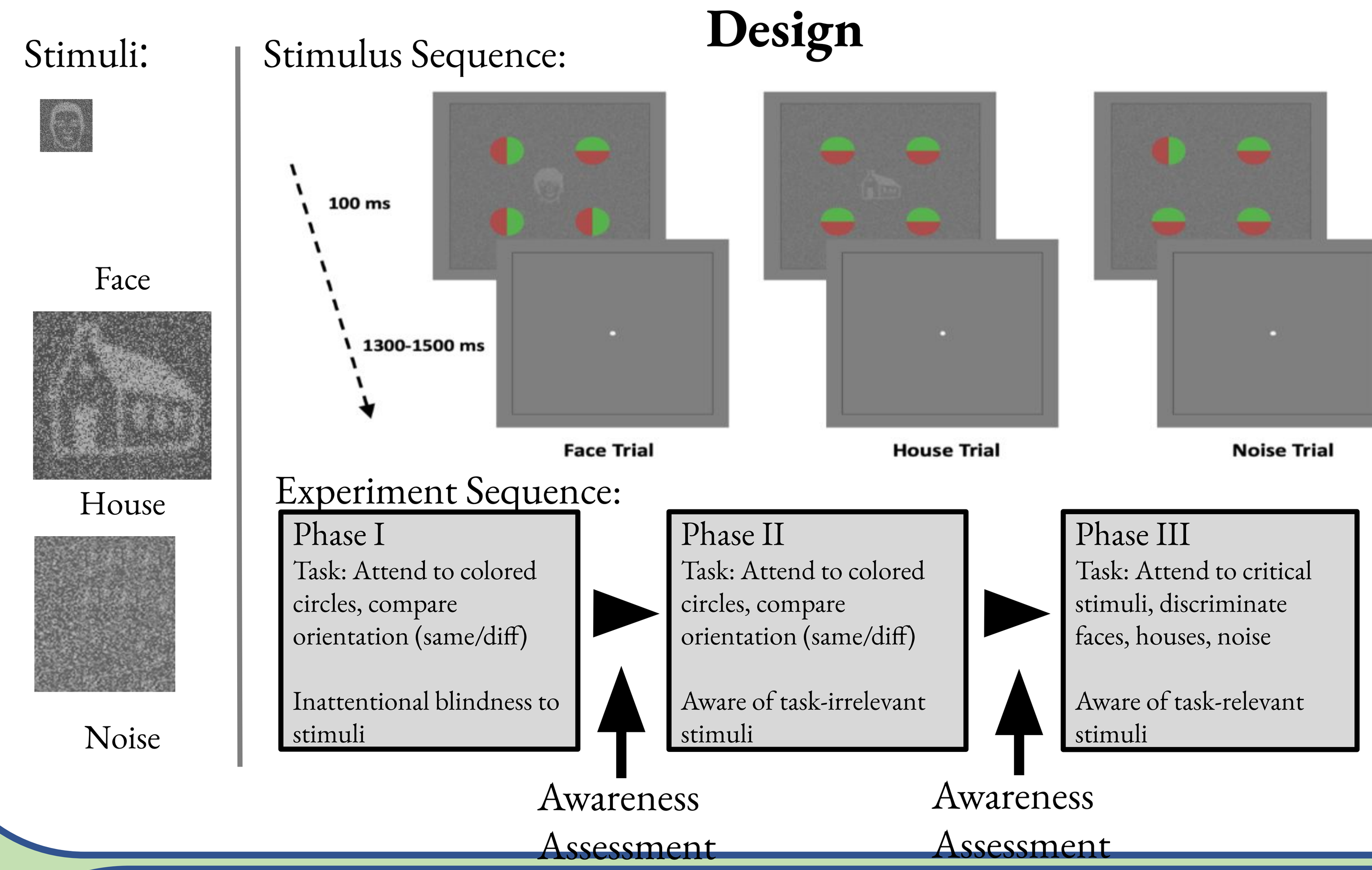

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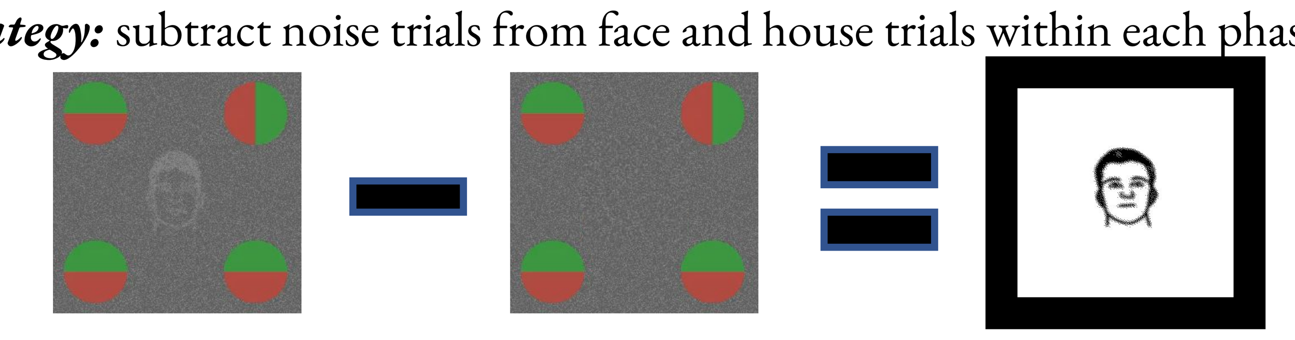

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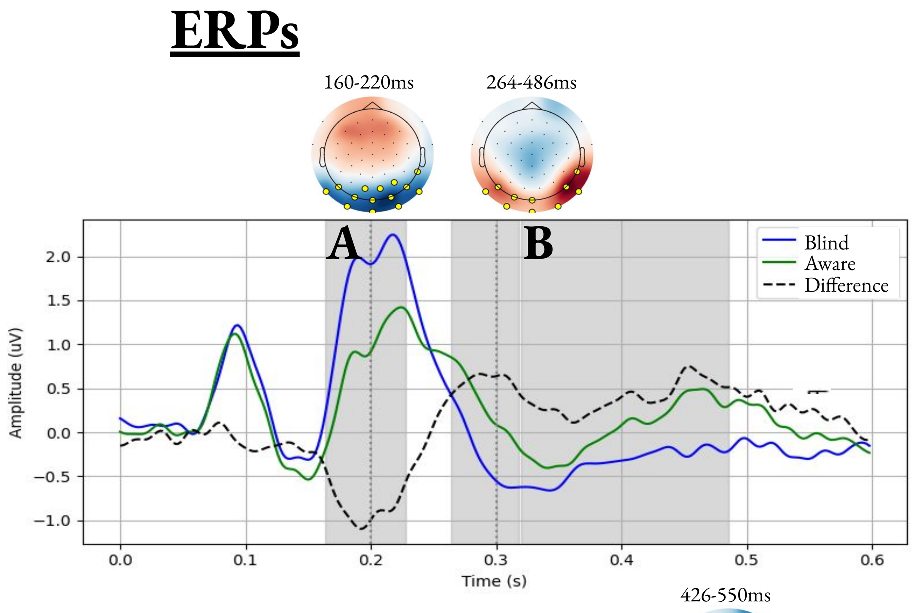

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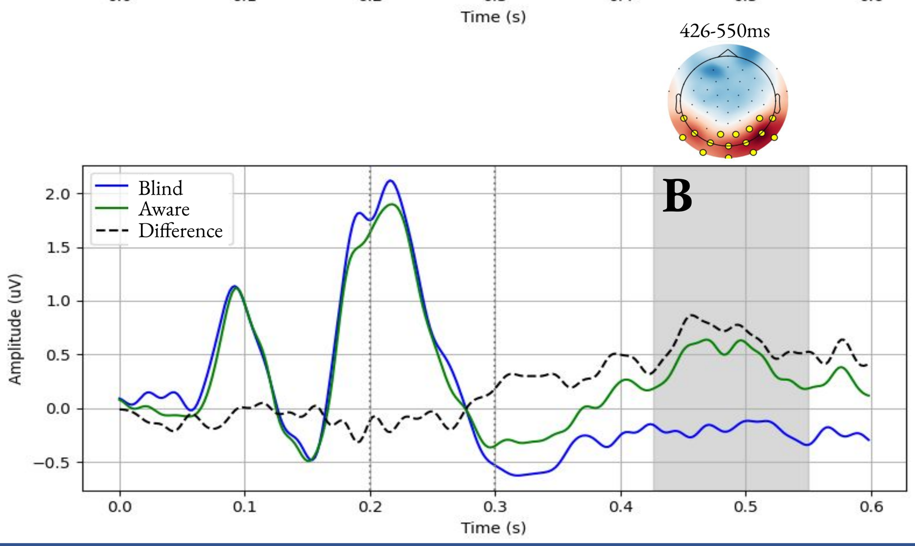

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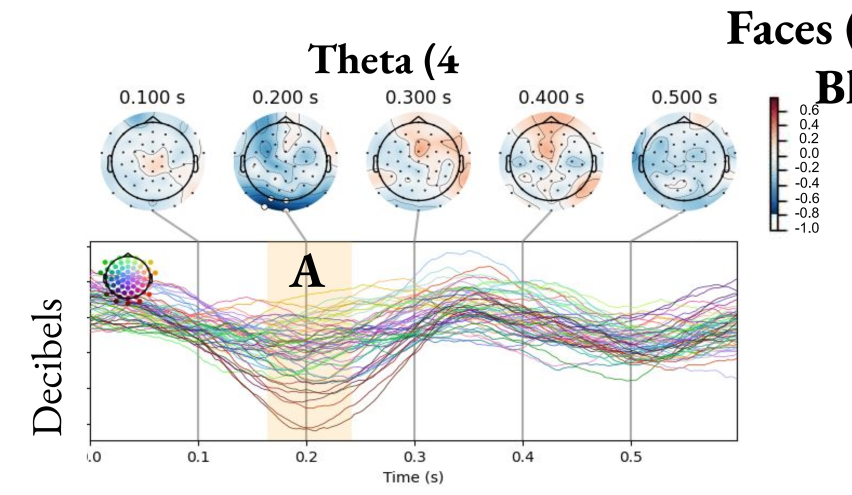

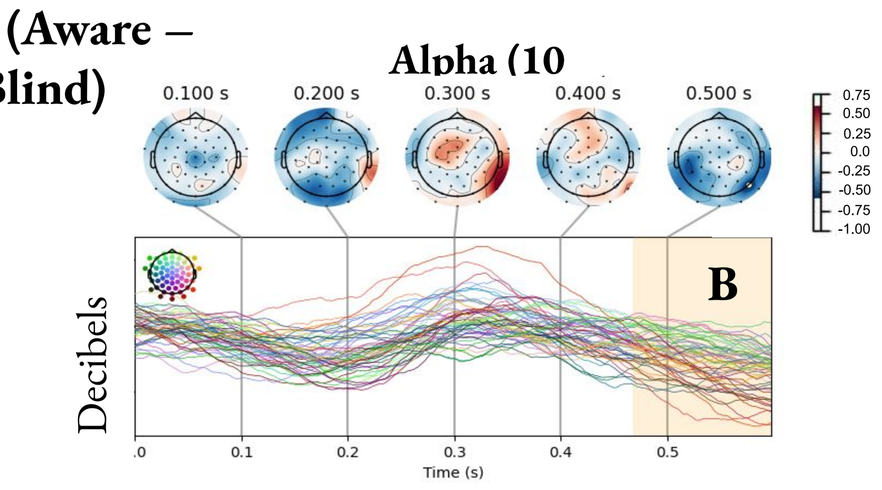

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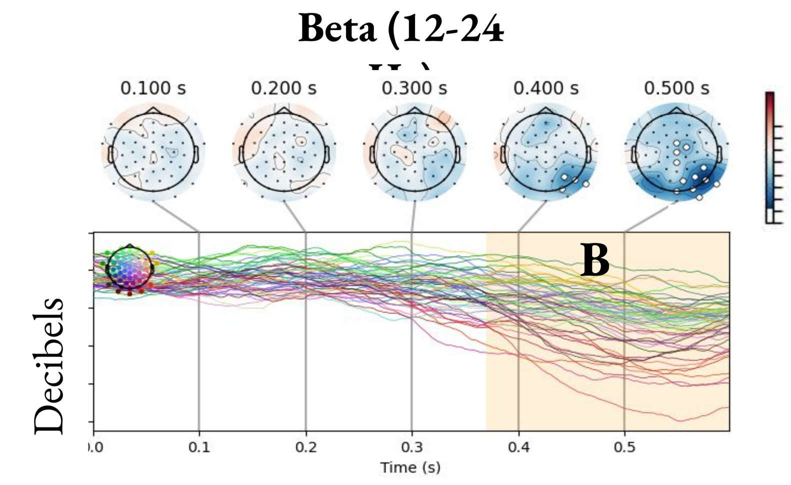

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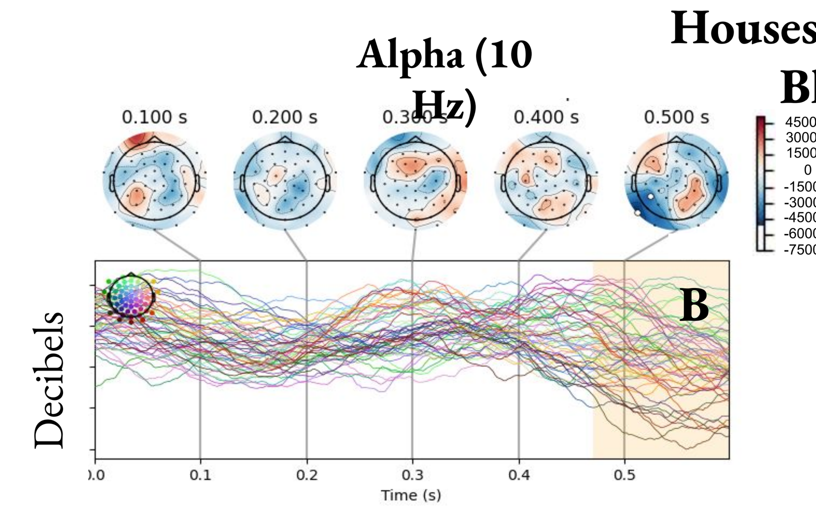

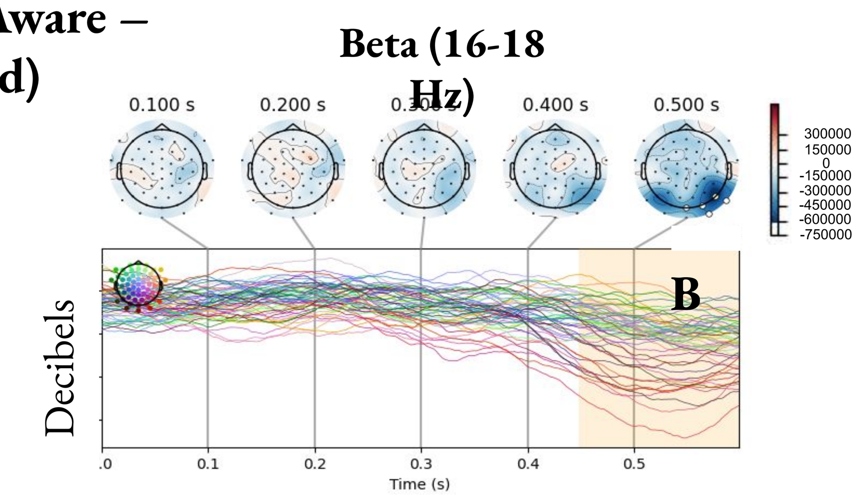

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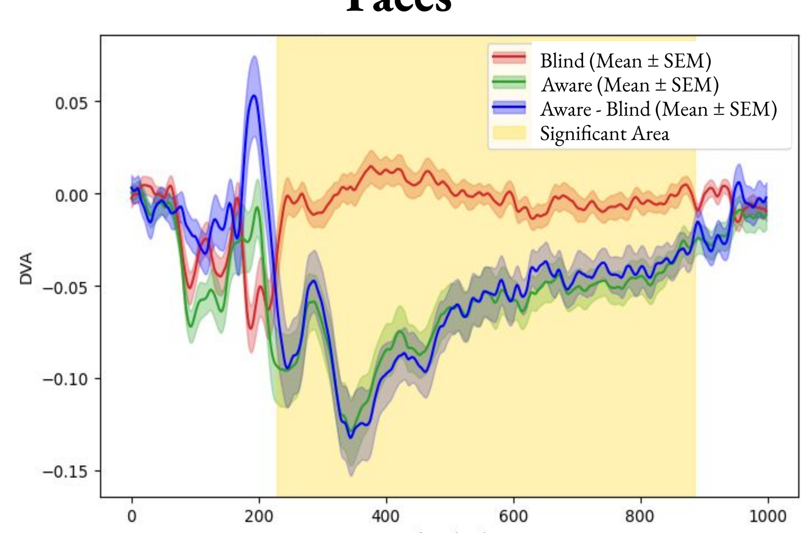

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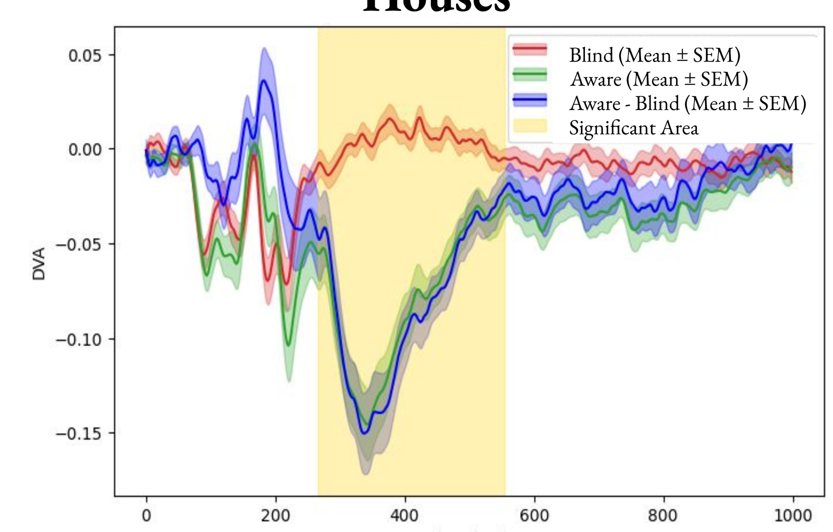

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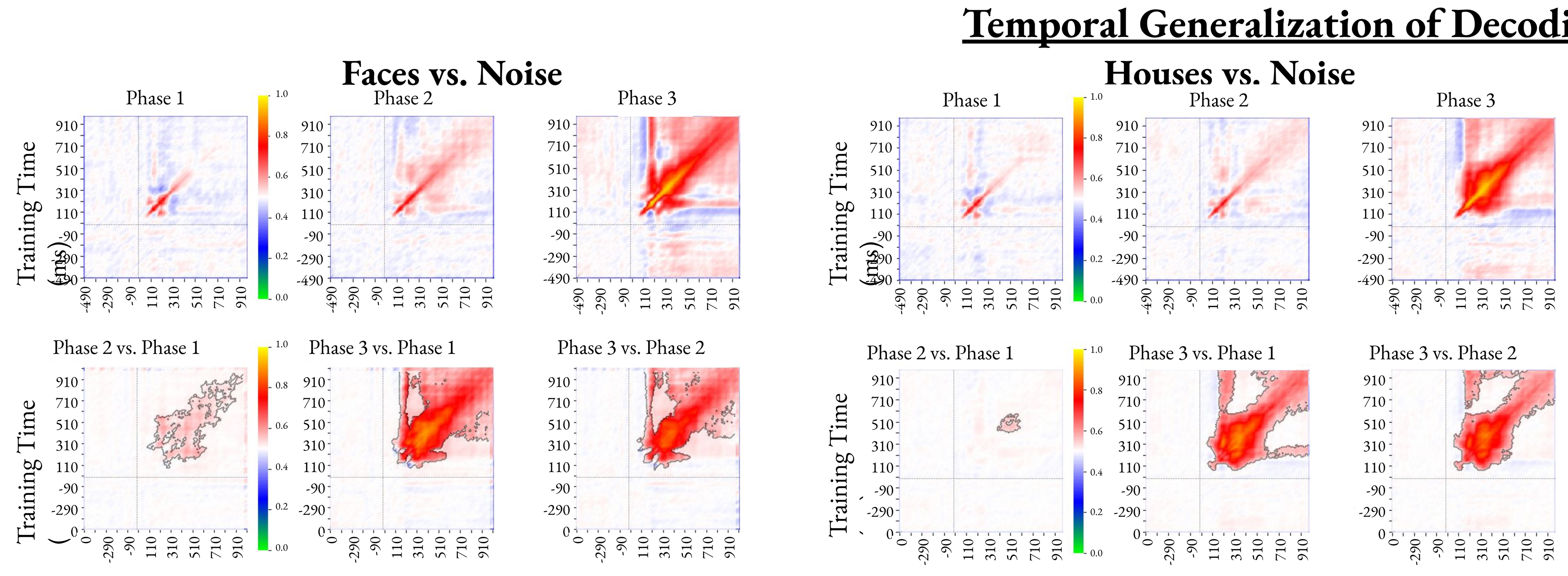

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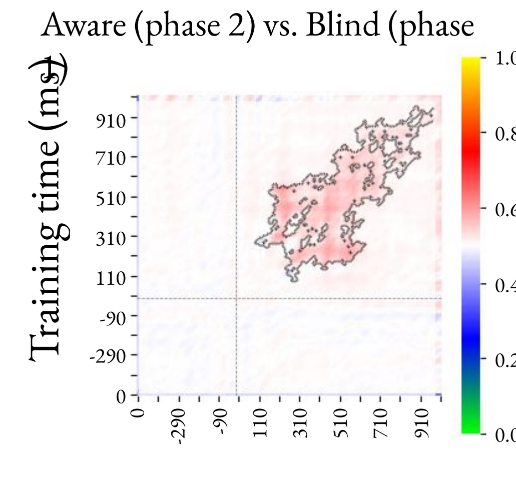

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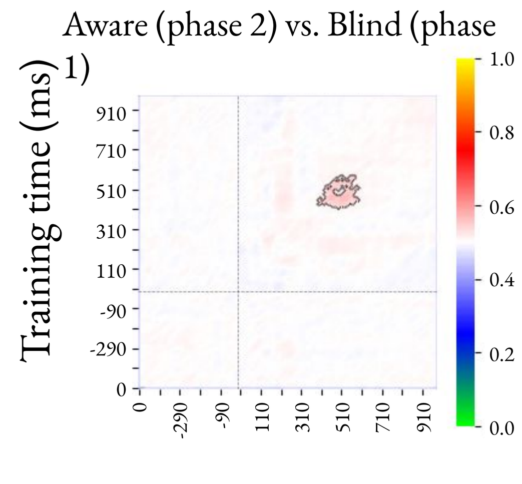

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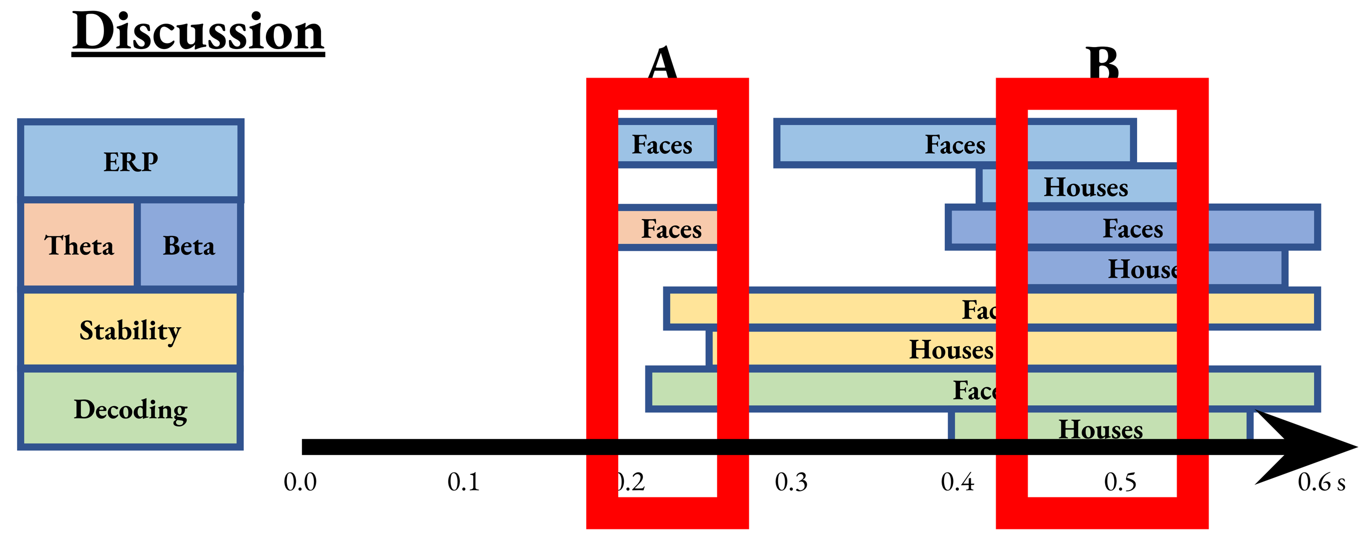

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